10PB24MATB03

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PRE-BOARD EXAMINATION 2024-25

CLASS 10 MATHS BASIC

Time allowed: 3 hours **Maximum marks: 80**

General instructions:

1. This question paper has five sections A, B, C, D & E.
2. Section A has 20 multiple choice questions carrying 1 mark each.
3. Section B has 5 questions carrying 2 marks each. Internal choice has been provided in 2 questions.
4. Section C has 6 questions carrying 3 marks each. Internal choice has been provided in 2 questions.
5. Section D has 4 questions carrying 5 marks each. Internal choice has been provided in 2 questions.
6. Section E has 3 Case Based integrated units of assessment (4 marks each) with sub – parts of the values of 1, 1 and 2 marks each.

Section A

1. The LCM of the smallest 2-digit number and the smallest composite number is

(A) 12 (B) 4 (C) 20 (D) 40

2. Find the largest number which divides 615 and 963 leaving remainder 6 in each case

(A) 18 (B) 81 (C) 58 (D) 87

3. The LCM of 2 numbers 180 and 288 is

(A) 1440 (B) 1140 (C) 1240 (D) 1340

4. If α & β are zeros of the polynomial $x^2 - 1$, then the value of $\alpha + \beta$ is

(A) 2 (B) 1 (C) -1 (D) 0

5. If a pair of linear equations in two variables is consistent, then the lines represented by the 2 equations are

(A) intersecting (B) parallel (C) always coincident (D) intersecting or coincident

6. Values of k for which the quadratic equation $2x^2 - kx - k = 0$ has equal roots

(A) 0 (B) 4 (C) 8   (D) 0 & −8

7. If x, 2x+9 , 4x+3 are three consecutive terms of an AP,then the value of x is

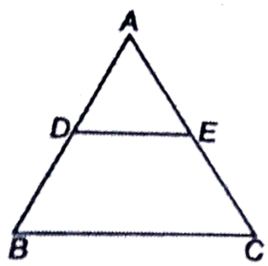
(A) 3   (B) 15   (C) 13   (D) 10

8. How many multiples of 4 lies between 10 and 250 ?

(A) 80   (B) 60 (C) 65 (D) 55

9. In the given figure $\frac{AD}{DB} = \frac{AE}{EC}$ and ∟ADE = 70° ,∟BAC = 50° , then ∟BCA =

(A) 70°   (B) 80°   (C) 50°   (D) 60°



10. If tangents PA and PB from a pont P to a circle with centre O are inclined to each other at an angle of 80° ,then∟POA is equal to

(A) 50°   (B) 20°   (C) 25°   (D) 30°

11. The angle of elevation of the top of a 15m high tower at a point 15m away from the base of the tower is

(A) 45°   (B) 30°   (C) 60° (D)90°

12. 2 $\cos^2 30 - 1$ is equal to

(A) sin 60°   (B) cos 60°   (C) tan 60°   (D) sec 60°

13.The area of quadrant of a circle whose circumference is 44 cm is

(A) $\frac{35}{2}$ cm^2 (B) $\frac{77}{2}$ cm^2.   C) $\frac{251}{2}$ cm^2.   (D) $\frac{3}{2}$ cm^2

14. Perimeter of a sector of a circle whose central angle is 90° and radius 7 cm is

(A) 35 cm (B)11 cm (C)22 cm   (D) 25cm

15. If cosec A – cot A = $\frac{4}{5}$,then cosec A =

(A) $\frac{47}{50}$ (B) $\frac{59}{40}$ (C) $\frac{51}{40}$ (D) $\frac{41}{40}$

16. If the perimeter and the area of a circle are numerically equal,then the radius of the circle is

(A) 2units (B)) 1unit (C)) 4 units (D)) 7 units

17 .The mean of of five observations x, x+2 , x+4, x+6 and x+8 is 11, then the value of x is

(A)4 (B)7 (C)11 (D) 6

18.There are 30 card of the same size in a bag in which the numbers 1 to 30 are written.One card is taken out of the bag at random.What is the probability that the number on the selected card is not divisible by 3

(A) $\frac{1}{15}$ (B)) $\frac{2}{3}$ (C)) $\frac{2}{15}$ (D)) $\frac{11}{15}$

Direction for questions 19 & 20: In question numbers 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

19**. Assertion:A quadratic polynomial having 4 and 3 as zeroes is $x^2 - 7x + 12$**

Reason: The quadratic polynomial having α & β as zeroes is given by $x^2 - (\alpha + \beta)x + \alpha\beta$

(a)Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)

(c) Assertion (A) is true but Reason (R) is false.

d) Assertion (A) is false but Reason (R) is true.

20. **Assertion**: D and E are points on the sides AB and AC respectively of a Δ ABC such that AD= 5.7cm, DB= 9.5cm AE=4.8cm and EC = 8cm. Then DE is not parallel to BC

Reason: If a line divides any two sides of a triangle in the same ratio, then it is parallel to the third side

(a)Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(b)Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)

(c)Assertion (A) is true but Reason (R) is false.

(d)Assertion (A) is false but Reason (R) is true.

Section B (5 questions of 2 marks each)

21. Find the ratio in which Y axis divides the line segment joining the points (5,−6) and (−1,−4).

OR

Point P (x,y) is equidistant from the points A (5.1) and B(1.5). Prove that x = y

22. Evaluate :

$\frac{sec^2 45° - \tan^2 45°}{\sin^2 45°}$.

23.The probability of guessing the correct answer to a certain test is $\frac{p}{12}$. If the probability of not guessing the correct answer to this question is $\frac{1}{3}$ .then find the value of p

24. Prove that a parallelogram circumscribing the circle is a rhombus

OR

Prove that the lengths of tangents drawn from an external point to a circle are equal.

25. Find the zeroes of the quadratic polynomial and verify the relation between zeroes and coefficients:

$2x^2 - 3 + 5x$

SECTION – C (6 questions of 3 mark each)

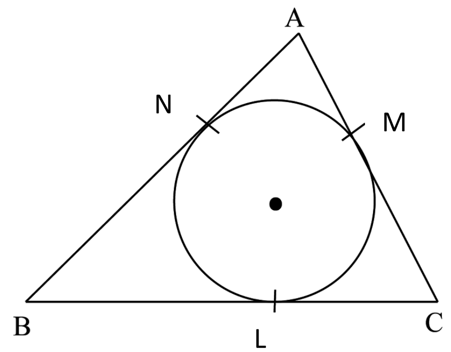
26.Prove that $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$

27. (a) Prove that $\sqrt{2}$ is an irrational number

OR

(b)Prove that $(\sqrt{3} - \sqrt{2})^2$ is irrational,given that $\sqrt{6}$ is an irrational number.

28.A circle is inscribed in a ΔABC having sides BC = 8cm ,AB = 10 cm and AC = 12cm. Find the lengths of BL, CM and AN



29 .(a) Find the sum of first 16 terms of an AP whose n^{th} term is given by a_n= 5n – 3

OR

(b) If 9^{th} term of an AP is zero, prove that its 29^{th} term is double of its 19^{th} term.

30.A sector of a circle with radius 6cm subtends an angle of 30° at the centre.

(i)find the length of the arc

(ii) the area of the corresponding major sector

31.One card is drawn from a well shuffled deck of 52 cards Find the probability of getting

(a) a face card

(ii) the queen of diamonds

(iii) a spade

SECTION – D (4 questions of 5 mark each)

32. Prove that if a line is drawn parallel to one side of a triangle to intersect the other 2 sides in distinct points, then the other 2 sides are divided in the same ratio

33.(a)A plane left 30 min later than its scheduled time and in order to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed.

OR

(b)The denominator of a fraction is one more than twice the numerator. If the sum of the fraction and its reciprocal is $2\frac{16}{21}$, then find the fraction.

34. (a)From a point P on the ground, the angle of elevation of the top of a 10 m tall building is 30°.A flag is hoisted at the top of the building and the angle of elevation of the top of the

flagstaff from P is 45° . Find the length of the flagstaff and the distance of the building from the point P.

OR

(b)The shadow of a tower standing on a level ground is found to be 40m longer when the Sun's altitude is 30° than when it is 60°. Find the height of the tower.

35.A solid is in the shape of a right circular cone surmounted on a hemisphere, the radius of each of them being 3.5 cm and the total height of the solid is 9.5 cm. Find the volume of the solid.

SECTION – E

Case study 1

36. Mr. Anil arranged a party for some friends. The expenses of the lunch are partly fixed and partly proportional to the number of guests. The expenses amount to Rs.650 for 7 guests and Rs.970 for 11 guests. Denote the fixed expenses by x and proportional expense per person as y and answer the following questions.

Based on the above information, answer the following questions.

(i) Form the pair of linear equations in two variables representing the above situation?

(ii) (a)What is the fixed expense for the party?

OR

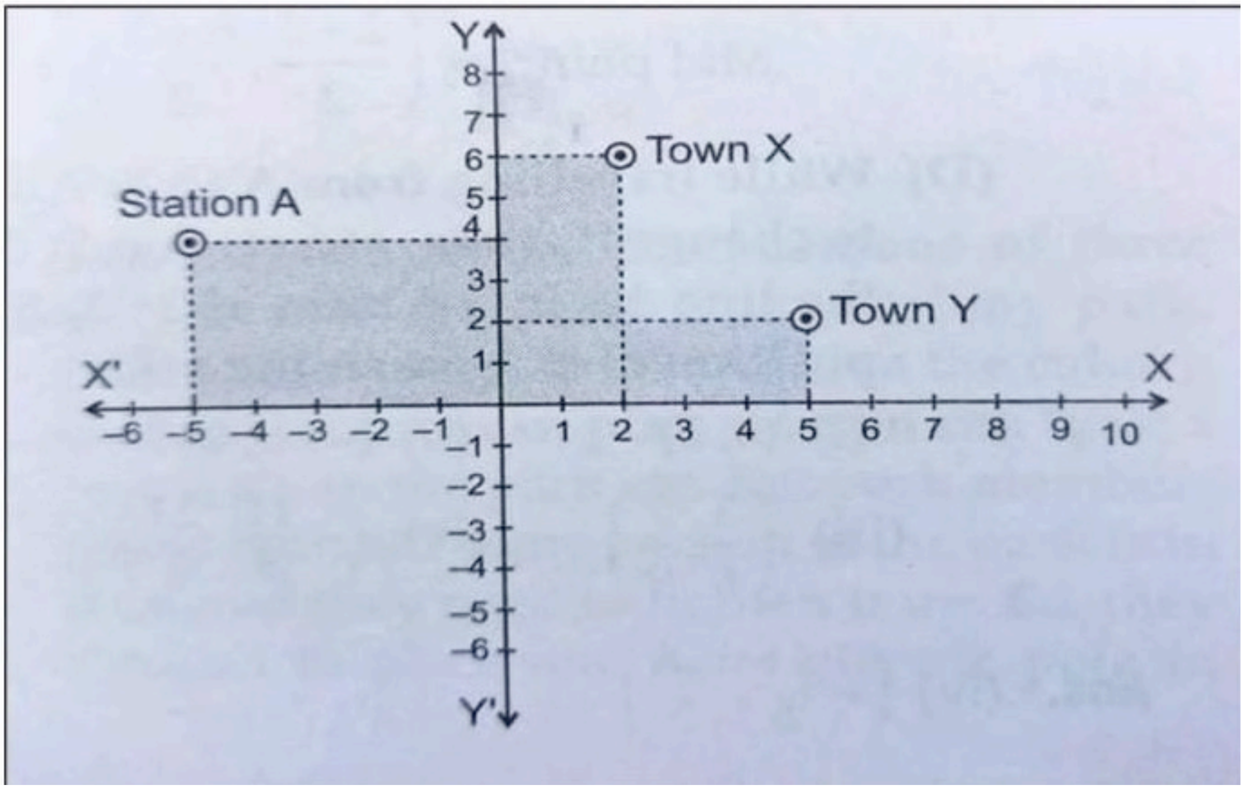
(b)Find the proportional expense per person (1+2+1)

(iii) If there would be 15 guests at the party, what amount Mr.Anil has to pay?

CASE STUDY II

37. Two friends Dalvin and Alice work in the same office in Toronto. In the Christmas vacation, they both decided to go to their home towns represented by Town X and Town Y. Town X and

Town Y are connected by trains from the same station A in Toronto. The position of Town X , Town Y and station A is shown on the coordinate axes.



Based on the given situation, answer the following questions:

i) What is the distance that Dalvin has to travel to reach his hometown X ?

ii) What is the distance that Alice has to travel to reach her hometown Y ?

iii) (a)While travelling from A to Y, Alice had to change the train, at a station, it divides the line AY in the ratio of 2:3, find the coordinates of position of that station.

OR (1+1+2)

iii) (b) In what ratio, Y - axis divides the line segment joining A and Y? Also find the coordinates of point of intersection.

CASE STUDY III

38. Life insurance is a contract between an insurance policy holder and an insurer or assurer, where the insurer promises to pay a designated beneficiary a sum of money upon the death of an insured person (often the policy holder). Depending on the contract, other events such as terminal illness or critical illness can also trigger payment. The policy holder typically pays a premium, either regularly or as one lump sum.

SBI life insurance agent found the following data for distribution of ages of 100 policy holders. Calculate the median age, if policies are given only to persons having age 18 years onwards but less than 60 years.

Age (in years)	Below 20	Below 25	Below 30	Below 35	Below 40	Below 45	Below 50	Below 55	Below 60
Number of policy holders	2	6	24	45	78	89	92	98	100

(i) Draw a frequency distribution table for the above data

(ii) (a)What is the median value of age?

OR

(ii) (b) What will be the upper limit of the modal class? What is the mode value of age?

(iii) Find the mean value of age using empirical relation. (1 +2+1)